

Use-Case Flow Specification

Use Case: Process Sales Checkout

Primary Actor: Pharmacy Clerk

Preconditions: The Pharmacy Clerk is authenticated and logged into the active checkout terminal. The system database contains accurate, up-to-date inventory counts and batch expiry dates.

Postconditions: The dispensed medicine quantity is successfully deducted from the specific batch's inventory record. The sales transaction is finalized and recorded in the system logs.

Main Success Scenario

1. The Pharmacy Clerk initiates a new sales checkout and inputs the requested medicine name or barcode.
2. The system queries the inventory database for available active stock.
3. The system enforces the FEFO (First-Expired-First-Out) picking order by identifying the valid batch with the nearest expiry date.
4. The system prompts the Pharmacy Clerk on the screen to physically retrieve that specific batch number.
5. The Pharmacy Clerk retrieves the item and confirms the correct batch on the terminal.
6. The system deducts the requested quantity from that specific batch's inventory count.
7. The system finalizes the checkout, records the transaction, and generates a dispensing label.

Alternate Flow

- 4a. Batch Expired or Damaged: The Pharmacy Clerk locates the requested FEFO batch but discovers it is physically damaged or has already expired on the shelf.
- 4a1. The Pharmacy Clerk selects "Mark Damaged/Expired" on the checkout prompt for that specific batch.
- 4a2. The system removes that batch from the active, sellable inventory.
- 4a3. The system recalculates the FEFO rule and identifies the next oldest valid batch in the system.
- 4a4. The system prompts the Clerk to retrieve the newly identified batch, and the flow returns to Step 5 of the Main Success Scenario.